

Predicting Hospital Performance in California: A Multi-Method Analytics Approach Using Linear and Logistic Regression

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Abstract

Hospital performance assessment is a critical component of healthcare quality management and policy planning. This study applies a full-stack data analytics pipeline to an empirical dataset of 137 California hospital clinical records drawn from the California Department of Health Care Access and Information (HCAI) 2016 Heart Failure inpatient cohort to examine factors associated with institutional performance ratings. Using Python (scikit-learn), R (tidyverse, ggplot2, caret), SQL (Google BigQuery), and business intelligence tools (Power BI, Tableau, Looker Studio), we conducted exploratory data analysis (EDA) and trained both linear and logistic regression models. The dataset encompassed three hospital types (General, Community, Teaching) across four geographic regions, with key predictors including risk-adjusted mortality rate, adverse event rate, and institutional type. The linear regression model yielded a test-set R^2 of 0.41 (CV $R^2 = 0.02$, SD = 0.20), suggesting moderate fit on the held-out test set but high fold-to-fold instability consistent with the multifactorial nature of real administrative clinical data. The logistic regression classifier achieved 64.3% accuracy and an AUC of 0.77 in distinguishing high-performing (Rating ≥ 70) from low-performing hospitals. Risk-adjusted mortality rate was the strongest negative predictor across both models, while Teaching hospital type emerged as the strongest positive predictor. Findings underscore the need for richer feature sets, larger longitudinal samples, and non-linear ensemble modeling approaches in real-world hospital quality research.

Keywords: hospital performance, regression analysis, healthcare analytics, HCAI, California hospital quality, scikit-learn, ggplot2, BigQuery ML, Heart Failure outcomes, Saint Louis University

1. Introduction

The quality and safety of hospital care are among the most pressing concerns in modern healthcare systems. Hospital performance ratings serve as composite proxies for care quality, patient safety, operational efficiency, and clinical outcomes (Jha et al., 2012). In the United States, initiatives such as the CMS Hospital Compare program and the Leapfrog Group's annual survey have generated increasing interest in identifying institutional predictors of superior outcomes (Chassin & Loeb, 2013).

California, with over 400 licensed general acute care hospitals serving the most populous state in the nation, represents a rich context for such inquiry. Prior studies have documented substantial variation in hospital quality across California, attributable to hospital type, geographic region, staffing levels, payer mix, and patient acuity (Needleman et al., 2011). Translating these findings into predictive models that support real-time decision-making remains an active research area.

The rapid evolution of data analytics tools — cloud SQL platforms (Google BigQuery), machine learning frameworks (scikit-learn, caret), statistical visualization (ggplot2), and interactive BI dashboards (Tableau, Power BI, Looker Studio) — has created unprecedented opportunities for end-to-end analytical pipelines (Chen et al., 2014). This paper addresses two primary research questions:

RQ1: Which hospital-level characteristics (type, region, risk-adjusted mortality rate, adverse event rate) are most strongly associated with overall performance rating?

RQ2: To what extent can a logistic regression model accurately classify hospitals as high performers (Performance Rating ≥ 70) versus low performers, using publicly available HCAI clinical data?

The complete project codebase — Python scripts, R scripts, SQL queries, and dashboard configuration files — is publicly available at <https://github.com/patelsahils9099-alt>.

2. Literature Review

2.1 Hospital Performance Measurement

The measurement of hospital performance has evolved significantly over three decades. Early frameworks focused narrowly on mortality rates and readmission frequencies (Iezzoni, 1997). Contemporary composite indices incorporate patient experience scores, process-of-care adherence, structural characteristics, and safety event rates into summary metrics (Shwartz et al., 2015). The Performance Rating used in this study reflects this composite philosophy, synthesizing clinical, operational, and safety dimensions into a single 0–100 score derived from the HCAI hospital quality registry.

Research consistently identifies hospital type as a significant predictor of clinical outcomes. Teaching hospitals exhibit superior outcomes for complex conditions, attributed to specialized staff, research infrastructure, and academic accountability (Burke et al., 2020). This study's empirical findings confirm that Teaching hospitals in the California HCAI dataset outperform General and

Community facilities by a substantial margin.

2.2 Risk Adjustment in Performance Measurement

Risk adjustment — controlling for patient severity, comorbidities, and demographics — is essential to ensure performance differences reflect institutional quality rather than patient mix (Iezzoni, 1997). The `Risk_Adjusted_Rate` variable in the HCAI dataset operationalizes this concept using AHRQ Patient Safety Indicator (PSI) software, which standardizes inpatient mortality rates for Heart Failure cases against statewide severity benchmarks. Failure to risk-adjust can introduce systematic biases disadvantaging hospitals serving higher-acuity populations (Werner & Bradlow, 2006).

2.3 Predictive Modeling in Healthcare Quality

Linear regression has been applied to predict continuous composite scores (Ash et al., 2012), while logistic regression has been used to classify facilities into quality tiers (Mukamel et al., 2008). Ensemble methods such as random forests have demonstrated superior predictive accuracy but at the cost of interpretability (Obermeyer & Emanuel, 2016). Google BigQuery ML enables training regression models directly in SQL without specialized programming environments (Google Cloud, 2023). The present study applies both model families to real HCAI administrative data to assess their practical utility.

2.4 Multi-Tool Analytics Workflows

The integration of tools such as BigQuery, Python, R, and Looker into unified pipelines has emerged as a dominant paradigm in data-driven organizations (Reis & Housley, 2022). In healthcare analytics, such stacks have been associated with improvements in reporting timeliness and analytic reproducibility (Simpao et al., 2014). The present study provides a documented, reproducible example of this architecture applied to real HCAI hospital quality data.

3. Methodology

3.1 Data Source and Description

The primary data analyzed in this study is sourced from the California Department of Health Care Access and Information (HCAI) hospital quality reporting registry, utilizing the 2016 Heart Failure inpatient clinical cohort. HCAI, formerly the Office of Statewide Health Planning and Development (OSHPD), publishes risk-adjusted inpatient mortality indicators annually for all licensed California acute care hospitals, computed using AHRQ Patient Safety Indicator (PSI) software. The dataset is publicly accessible at the California Health and Human Services Open Data Portal (data.chhs.ca.gov). Heart Failure was selected as the index condition because it is reported by the largest number of facilities across all hospital types and regions in California.

After excluding statewide aggregate rows and hospitals with missing Heart Failure data, the analytical sample comprised 137 California hospitals spanning 30 counties across four geographic regions. Each record contains the hospital name, official OSHPD facility identifier (OSHPDID), county, risk-adjusted mortality rate, number of inpatient deaths, number of Heart Failure cases, and HCAI quality rating category (Better, As Expected, Worse). A composite Performance_Rating (0–100) was derived from the HCAI rating category and risk-adjusted mortality rate to support continuous regression modeling. All statistics reported in this paper are computed directly from the source data using Python (pandas). Raw data and scripts are available at <https://github.com/patelsahils9099-alt>.

Table 1 California HCAI Heart Failure Dataset — Variables and Descriptive Statistics (n = 137)

Variable	Type	Range / Values	Mean	SD
Performance_Rating	Continuous	27.6 – 99.6	71.75	13.61
Risk_Adjusted_Rate	Continuous	0.0 – 18.9%	3.12	1.87
Cases	Integer	11 – 2,103	459.2	294.5
Adverse_Events	Integer	0 – 42	14.24	7.98
Hospital_Type	Categorical	General (62.0%), Community (31.4%), Teaching (6.6%)	—	—
Region	Categorical	Southern CA (36.5%), Bay Area (28.5%), Central Valley (18.2%), Northern CA (16.8%)	—	—
Adverse_Rate*	Engineered	0.000 – 0.190	0.034	0.019
High_Performer*	Binary	1 if Rating ≥ 70; else 0 (56.2% High, 43.8% Low)	—	—

Note. * = engineered features. Adverse_Rate = Adverse_Events / Cases. High_Performer threshold = 70 (Performance_Rating sample median = 72.9). Source: California HCAI Inpatient Mortality Indicators, Heart Failure cohort, 2016.

3.2 Data Cleaning and Feature Engineering

Data were loaded using pandas. Missing value analysis confirmed zero missing entries across all analytical variables. Zero duplicate hospital records were found. Categorical variables were standardized to title case. One-hot encoding was applied to Hospital_Type and Region with Community (Hospital_Type) and Central Valley (Region) retained as implicit reference categories by exclusion from the feature matrix.

Two feature engineering decisions warrant explicit justification. First, the County column (30 unique California counties) was excluded from the regression feature matrix. Including 30 binary county indicators in a model with $n = 137$ observations would result in severe overparameterization — approximately one free parameter for every 4.6 observations — dramatically increasing variance and producing unstable, uninterpretable coefficient estimates. Region (four categories) was retained as a theoretically grounded, lower-cardinality geographic proxy. Second, Adverse_Events and Cases were excluded individually; instead, their ratio ($\text{Adverse_Rate} = \text{Adverse_Events} / \text{Cases}$) was retained as the normalized adverse burden measure. Including all three simultaneously produced multicollinearity with inflated opposing coefficients ($\text{VIF} > 10$), confirming their statistical redundancy.

3.3 Exploratory Data Analysis

EDA was conducted in Python (matplotlib, seaborn) and R (ggplot2, corrplot), including: (a) histograms and boxplots of all continuous variables; (b) group comparisons across Hospital_Type and Region; (c) bivariate scatter plots with OLS trend lines; and (d) a Pearson correlation matrix. SQL aggregation queries in Google BigQuery replicated all group summaries for cloud deployment validation.

Interactive dashboards were constructed in Power BI Desktop and Tableau Desktop, with charts including KPI cards, regional heat maps, scatter plots with regression trend lines, and performance distribution histograms. The Tableau workbook is published and publicly accessible on Tableau Public at public.tableau.com/app/profile/sahil.patel. Looker Studio was connected directly to the BigQuery feature view for live-refresh web reporting.

3.4 Statistical Models

3.4.1 Linear Regression

A multiple linear regression model was estimated with Performance_Rating as the continuous outcome. The feature matrix comprised seven variables: Risk_Adjusted_Rate, Adverse_Rate, and five binary indicators (two for Hospital_Type — General and Teaching — and three for Region — Bay Area, Northern California, and Southern California). All features were z-score normalized (StandardScaler, scikit-learn). Performance was evaluated on a 20% held-out test set ($n = 28$) via R^2 , MAE, and RMSE, and internally validated via five-fold cross-validation. An equivalent model was trained in BigQuery ML (`model_type = 'LINEAR_REG'`) and in R via `lm()`.

3.4.2 Logistic Regression

A binary logistic regression model with High_Performer (0/1) as the outcome was estimated using L2-regularized LogisticRegression (scikit-learn, $C = 1.0$, $\text{max_iter} = 1,000$). Evaluation metrics included accuracy, precision, recall, F1, confusion matrix, AUC-ROC, and five-fold cross-validated accuracy. Odds ratios were computed by exponentiating standardized coefficients. An equivalent model was estimated in R using `glm()` with `family = binomial(link = 'logit')` and in BigQuery ML (`model_type = 'LOGISTIC_REG'`).

3.5 Software and Reproducibility

All analyses used Python 3.10 (pandas 2.0, scikit-learn 1.3, matplotlib 3.7, seaborn 0.12), R 4.3.0 (tidyverse 2.0, ggplot2 3.4, caret 6.0, pROC 1.18), Google BigQuery Standard SQL with BigQuery ML, Power BI Desktop, Tableau Desktop 2024.1, and Looker Studio. All code is version-controlled and publicly accessible at <https://github.com/patelsahils9099-alt>.

4. Results

4.1 Descriptive and Exploratory Findings

The 137-hospital California HCAI sample yielded a mean Performance_Rating of 71.75 (SD = 13.61, median = 72.9, range: 27.6–99.6), exhibiting moderate left skew. General hospitals formed the largest group (n = 85, 62.0%), followed by Community (n = 43, 31.4%), and Teaching (n = 9, 6.6%). Regionally, Southern California contributed the most hospitals (n = 50, 36.5%), followed by Bay Area (n = 39, 28.5%), Central Valley (n = 25, 18.2%), and Northern California (n = 23, 16.8%). Of the 137 hospitals, 77 (56.2%) were classified as High Performers (Performance_Rating $\geq$ 70) and 60 (43.8%) as Low Performers. The HCAI quality rating distribution was: As Expected (n = 95, 69.3%), Better (n = 33, 24.1%), Worse (n = 9, 6.6%).

Table 2 Mean Performance Rating by Hospital Type — California HCAI Dataset (n = 137)

Hospital Type	n	% of Sample	Mean Rating	SD
Teaching	9	6.6%	85.57	11.24
General	85	62.0%	71.51	13.88
Community	43	31.4%	69.32	12.41

Teaching hospitals substantially outperformed General and Community facilities (mean = 85.57 vs. 71.51 vs. 69.32), consistent with prior literature on academic medical centre superiority for complex conditions (Burke et al., 2020). Regional differences favored Central Valley (M = 73.67) and Bay Area (M = 73.21) over Southern California (M = 72.60), with Northern California recording the lowest mean performance (M = 65.33, SD = 14.82). Risk-adjusted mortality rates were right-skewed (mean = 3.12%, range: 0.0–18.9%), reflecting the well-documented heterogeneity of Heart Failure outcomes across California's hospital landscape.

Table 3 Mean Performance Rating by Region — California HCAI Dataset (n = 137)

Region	n	Mean Rating	SD	HCAI Avg. Mortality Rate
Central Valley	25	73.67	13.11	3.02%
Bay Area	39	73.21	13.44	2.91%
Southern California	50	72.60	14.22	3.18%
Northern California	23	65.33	14.82	3.41%

Note. Performance differences across regions were directionally consistent with the logistic regression coefficients. All Pearson correlations among numeric predictors and the outcome were modest in magnitude ($|r| \leq 0.42$), consistent with the multifactorial nature of real administrative clinical data.

4.2 Linear Regression Results

The multiple linear regression model produced a test-set R^2 of 0.41 (MAE = 9.16, RMSE = 10.94, n_{test} = 28). While this suggests the model explains approximately 41% of Performance_Rating variance on the held-out test partition, five-fold cross-validation revealed high fold-to-fold instability (CV R^2 = 0.02, SD = 0.20; fold values: -0.001, -0.351, 0.102, 0.229, 0.105). This pattern — reasonable test-set fit but low average cross-validated R^2 — is characteristic of small clinical datasets (n = 137) where the composition of training and test partitions materially affects model performance. It confirms that the seven available features carry some predictive signal but are insufficient for robust generalization, motivating the collection of richer administrative covariates. Risk_Adjusted_Rate was the strongest coefficient (β = -8.75), confirming that hospitals with higher risk-adjusted Heart Failure mortality are rated lower — the most clinically meaningful finding of the linear analysis.

Table 4 Linear Regression — Test Set Performance, California HCAI Data (n_{test} = 28)

Metric	Value	Interpretation
R^2 (Test Set)	0.41	41% of outcome variance explained on held-out set
MAE	9.16	Average absolute error in rating-scale points
RMSE	10.94	Root mean squared error
CV R^2 (5-fold mean)	0.02	Low average cross-validated fit; high fold variance
CV R^2 SD	0.20	High instability — reflects small sample sensitivity
Training n	109	80% of 137 hospitals used for fitting

Table 5 Linear Regression — Standardised Coefficients (All Features, Sorted by $|\beta|$)

Feature	Coefficient (β)	Direction / Interpretation
Risk_Adjusted_Rate	-8.75	Strongest predictor: higher mortality → lower rating
Type_Teaching	+3.17	Teaching hospitals score significantly higher
Adverse_Rate	+2.20	Positive direction; interpret with caution (collinearity risk)
Region_Northern CA	-1.74	Northern CA associated with lower ratings
Region_Southern CA	-1.34	Southern CA negative vs. Central Valley reference
Region_Bay Area	-1.21	Bay Area negative vs. Central Valley reference
Type_General	+0.27	Weak positive vs. Community reference

4.3 Logistic Regression Results

The logistic regression classifier achieved 64.3% accuracy on the test set (n = 28), with AUC-ROC = 0.77 — indicating good discriminative ability, substantially above the 0.50 chance baseline. Five-fold cross-validation confirmed generalizability (CV Accuracy = 70.8%, SD = 6.9%).

The model demonstrated higher recall for the High Performer class (0.75) than for Low Performers (0.50), reflecting the modest class imbalance (56.2% High, 43.8% Low). The confusion matrix (6 true negatives, 6 false positives, 4 false negatives, 12 true positives) indicates the model is more effective at confirming high performers than at ruling out low performers.

Table 6 Logistic Regression — Classification Report, California HCAI Data (n_test = 28)

Class	Precision	Recall	F1	Support
Low Performer (0)	0.60	0.50	0.55	12
High Performer (1)	0.67	0.75	0.71	16
Macro Average	0.63	0.62	0.63	28
Weighted Average	0.64	0.64	0.64	28

Table 7 Logistic Regression — Standardised Coefficients and Odds Ratios, California HCAI Data

Feature	Coefficient	Odds Ratio	Interpretation
Type_Teaching	+0.70	2.01	Teaching hospitals twice as likely to be High Performers
Risk_Adjusted_Rate	−1.19	0.30	Strongest negative: higher mortality → 70% lower odds
Adverse_Rate	−0.42	0.66	Higher adverse event burden → lower performance odds
Region_Northern CA	−0.57	0.57	Northern CA hospitals lower odds vs. Central Valley
Region_Southern CA	−0.32	0.73	Southern CA slight negative vs. Central Valley
Region_Bay Area	−0.76	0.47	Bay Area lower odds — reflects case-mix complexity
Type_General	−0.01	0.99	Negligible difference from Community reference

5. Discussion

5.1 Linear Regression: Real Data Complexity and Model Instability

The linear model's test-set R^2 of 0.41 is a moderately encouraging result for an administrative dataset with only seven predictors. However, the low and highly variable cross-validated R^2 (mean = 0.02, SD = 0.20) reveals an important limitation of small real-world clinical datasets: model performance is sensitive to which hospitals happen to fall in the training versus test partition. This is not a methodological failure but an empirical confirmation of what the literature has long argued: composite hospital performance scores are driven by a complex array of structural, cultural, and patient-level factors well beyond what standard administrative datasets can capture (Shwartz et al., 2015). Variables such as nursing staffing ratios, physician board certification rates, EHR adoption, payer mix, and community socioeconomic context — all documented predictors — were unavailable in the HCAI Heart Failure registry. With only seven features and $n = 137$, the model is statistically underpowered for robust generalization. The finding that Risk_Adjusted_Rate carries the largest negative coefficient ($\beta = -8.75$) is nevertheless highly interpretable and clinically meaningful: hospitals with higher risk-adjusted Heart Failure mortality are rated systematically lower, exactly as the HCAI quality framework intends.

5.2 Logistic Regression: Meaningful Discrimination with Real Clinical Data

The AUC of 0.77 indicates that the logistic model can rank California hospitals by their High Performer probability substantially better than chance — a result that improves meaningfully on naive baselines (Hosmer & Lemeshow, 2013). The cross-validated accuracy of 70.8% further demonstrates that this discriminative ability generalizes across data folds. Teaching hospital type emerged as the strongest positive predictor (OR = 2.01): Teaching hospitals are approximately twice as likely to be classified as High Performers, consistent with robust evidence on the academic medical centre advantage (Burke et al., 2020). Risk_Adjusted_Rate carried the strongest negative association (OR = 0.30), confirming that higher inpatient mortality dramatically reduces the probability of high-performer classification. The unexpected negative coefficient for Bay Area hospitals (OR = 0.47) relative to the Central Valley reference is counterintuitive and warrants investigation; it may reflect that large Bay Area medical centres, while prestigious, treat more complex and high-acuity Heart Failure cases than their rural Central Valley counterparts, depressing their HCAI quality ratings despite high resource capability (Werner & Bradlow, 2006).

5.3 BigQuery ML and Multi-Tool Scalability

The parallel execution of regression models across scikit-learn, R's `glm()`, and BigQuery ML demonstrated consistent directional results across platforms. The BigQuery ML implementation enables real-time scoring of new hospital records as HCAI data is ingested into the cloud warehouse, without maintaining a separate Python serving environment — a meaningful operational advantage for health system analytics teams (Google Cloud, 2023). The Looker Studio dashboard connected directly to the BigQuery feature view provides hospital administrators and policy researchers with auto-refreshing visualisations accessible via any web browser.

5.4 Limitations

Key limitations of this study include: (1) the dataset is cross-sectional, limited to the fiscal year 2016, and focuses strictly on the Heart Failure inpatient cohort — findings may not generalize to other conditions or years; (2) the feature set is narrow, omitting staffing ratios, payer mix, accreditation status, and community socioeconomic context, all documented predictors of hospital quality; (3) the sample of 137 hospitals limits statistical power and cross-validated stability; (4) linear and logistic regression assume additive relationships — ensemble models (random forests, XGBoost) may capture non-linear interactions missed here; and (5) the High Performer threshold of 70 is empirically motivated by the sample median but is potentially arbitrary. All code, data, and model outputs are publicly archived at <https://github.com/patelsahils9099-alt> to facilitate replication and extension.

6. Conclusion

This study applied a comprehensive, multi-tool analytics pipeline to 137 California hospitals drawn from the HCAI 2016 Heart Failure inpatient registry — a real, publicly available clinical dataset with official OSHPD facility identifiers. The linear regression model achieved a test-set R^2 of 0.41, with high cross-validated instability ($CV\ R^2 = 0.02$) reflecting the inherent complexity of real administrative health data. The logistic classifier achieved $AUC = 0.77$ and 70.8% cross-validated accuracy, demonstrating meaningful and generalizable binary discrimination.

Risk-adjusted Heart Failure mortality rate and Teaching hospital type were the most consistent, clinically interpretable predictors across both models. From a methodological standpoint, this study provides a fully reproducible template for hospital analytics projects integrating Python, R, SQL, BigQuery ML, and BI tools using real government health data — all code is open-source at <https://github.com/patelsahils9099-alt>.

Future work should incorporate richer feature sets (staffing ratios, payer mix, EHR adoption), apply ensemble models (random forests, gradient boosting), extend the analysis longitudinally across HCAI's 2016–2023 multi-year dataset, and validate findings across additional clinical condition cohorts. As healthcare organisations increasingly adopt cloud-native platforms, the workflow demonstrated here — anchored in real public-sector data — represents a viable and reproducible model for evidence-based hospital quality management in California and beyond.

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